MULTIPLE LINEAR REGRESSION

Here we have a data set,

X_1	X_2	---	X_p	Y
x_{11}	x_{21}	---	x_{p1}	y_1
x_{12}	x_{22}	---	x_{p2}	y_2
⋮	⋮		⋮	⋮
x_{1n}	x_{2n}	---	y_{pn}	y_n

In LAD regression we minimise

$$\sum |y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - \cdots - \beta_p x_{pi}|$$

(i) Iterative method.

$$\sum |\underbrace{y_i - \beta_1^{(0)} x_{1i} - \beta_2^{(0)} x_{2i} \cdots - \beta_p^{(0)} x_{pi}}_{y_i^{*}} - \beta_0|$$

and this equals $\sum |y_i^{*} - \beta_0|$ and $\beta_0^{(0)}$ will be the median of y_i^{*} and $\beta_0^{(1)}$ we will get

now update $\beta_1^{(0)}$ with $\beta_1^{(1)}$

$$\sum |y_i - \beta_2^{(0)} x_{2i} - \beta_3^{(0)} x_{3i} - \cdots - \beta_p^{(0)} x_{pi} - \beta_0^{(0)} - x_{1i}\beta_1^{(1)}|$$

$\beta_1^{(1)}$ will be weighted median of $\sum |x_{1i}| \left| \frac{y_i}{x_{1i}} - \beta_2^{(0)} \frac{x_{2i}}{x_{1i}} - \beta_3^{(0)} \frac{x_{3i}}{x_{1i}} - \cdots - \frac{\beta_0}{x_{1i}} - \beta_1 \right|$

$|x_{1i}|\ \ y_1^{*}$

Now go on updating $\beta_2^{(1)}, \beta_3^{(1)} \cdots \beta_p^{(1)}$ in this way.

Now we get $\beta_1^{(1)}, \beta_2^{(1)}, \beta_3^{(1)} \cdots \beta_p^{(1)}$ and then find $\beta_0^{(1)}$ and this will go on and either we have limit on $\sum |y_i - \hat{y}_i| < \epsilon$ or no. of iterations.

(ii) Iteratively re weighted least square (IRWLS)

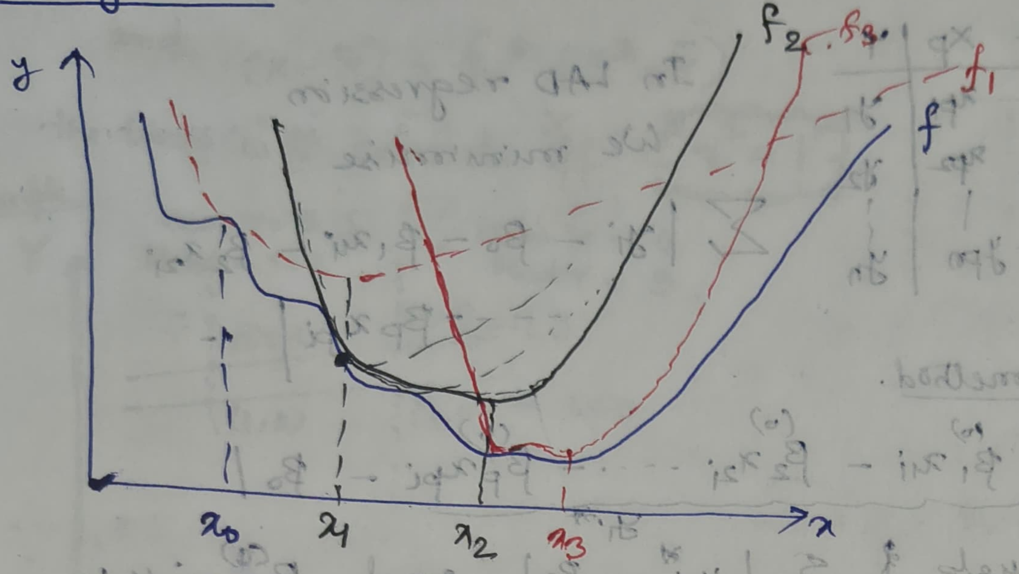
Start with $\beta_0^{(0)}, \beta_1^{(0)}, \cdots \beta_p^{(0)}$ initial choices for $\beta_0, \beta_1, \cdots \beta_p$ and minimize

$$\sum |y_i - \beta_0 - \beta_1 x_{1i} - \cdots - \beta_p x_{pi}| \qquad \left[\because |x| = \frac{x^2}{|x|}\right]$$

$$\sum \frac{(y_i - \beta_0 - \beta_1 x_{1i} - \cdots - \beta_p x_{pi})^2}{|y_i - \beta_0^{(0)} - \beta_1^{(0)} x_{1i} - \cdots - \beta_p^{(0)} x_{pi}|} = \sum w_i^{(0)} [y_i - \beta_0 - \beta_1 x_{1i} - \cdots - \beta_p x_{pi}]^2$$

and after minimizing we will get $\beta_0^{(1)}, \beta_1^{(1)} - - \beta_p^{(1)}$.

MM algorithm



(i) $f_1(x) \geq f(x) \ \forall x$

(ii) $f_1(x_0) = f(x_0)$

$\Rightarrow$ then this f_1 is called majorization function for f.

Let x_1 be the minimizer for f_1

Now construct f_2 s.t

$f_2(x) \geq f(x) \ \forall x$

$f_2(x_1) = f(x_1)$

Minimize f_2 to get x_2

Now construct f_3 s.t

$f_3(x) \geq f(x) \ \forall x$

$f_3(x_2) = f(x_2)$

Minimize f_3 to get x_3 and proceed this way.

This algorithm is known as MM algorithm

$\hookrightarrow$ Majorization + Minimization

For converging x_0, x_1, x_2, x_3 ——

$f(x_0) \geq f(x_1) \geq f(x_2) \geq f(x_3)$

We have $f(x_0) = f_1(x_0) \geq f_1(x_1) \geq f(x_1)$

$\because$ x_1 is the minimizer of f_1

16/9/25

Let $f(\beta_0, \beta_1, \ldots \beta_p) = \sum \left| y_i - \beta_0 - \beta_1 x_{1i} - \cdots - \beta_p x_{pi} \right|$

In IRWLS, we started with a trial sol^n $\beta_0^{(0)}, \beta_1^{(0)}, \ldots \beta_p^{(0)}$ and minimize

$$g_0(\beta_1, \beta_2, \ldots \beta_p) = \sum \frac{(y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - \cdots - \beta_p x_{pi})^2}{\left| y_i - \beta_0^{(0)} - \beta_1^{(0)} x_{1i} - \cdots - \beta_p^{(0)} x_{pi} \right|}$$

Define

$$m_i = \frac{\left| y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - \cdots - \beta_p x_{pi} \right|}{\left| y_i - \beta_0^{(0)} - \beta_1^{(0)} x_{1i} - \beta_2^{(0)} x_{2i} - \cdots - \beta_p^{(0)} x_{pi} \right|}$$

$$f(\beta_0, \beta_1, \ldots \beta_p) = \sum m_i \left| y_i - \beta_0^{(0)} - \beta_1^{(0)} x_{1i} - \beta_2^{(0)} x_{2i} - \cdots - \beta_p^{(0)} x_{pi} \right|$$

$$g_0(\beta_0, \beta_1, \ldots \beta_p) = \sum m_i^2 \left| y_i - \beta_0^{(0)} - \beta_1^{(0)} x_{1i} - \beta_2^{(0)} x_{2i} - \cdots - \beta_p^{(0)} x_{pi} \right|$$

we know $(m_i - 1)^2 \geq 0$

$$\Rightarrow m_i \leq \frac{1}{2}(m_i^2 + 1) \quad \forall i = 1, 2, \ldots, n$$

$$\Rightarrow \sum_{i=1}^{n} m_i \left| y_i - \beta_0^{(0)} - \cdots - \beta_p^{(0)} x_i \right| \leq \frac{1}{2} \sum_{i=1}^{n} (m_i^2 + 1) \left| y_i - \beta_0^{(0)} - \beta_1^{(0)} x_{1i} - \cdots - \beta_p^{(0)} x_{pi} \right| \quad \text{--- (i)}$$

$$\Rightarrow f(\beta_0, \beta_1, \ldots \beta_p) \leq \underbrace{\frac{1}{2} g_0(\beta_0, \beta_1, \ldots \beta_p) + \frac{1}{2} \sum \left| y_i - \beta_0^{(0)} - \beta_1^{(0)} x_{1i} - \cdots - \beta_p^{(0)} x_{pi} \right|}_{\text{majorization func}^c}$$

and $f = g$ at $m_i = 1$ --- (ii)

ind of $\beta_0, \beta_1, \ldots \beta_p$

Minimization of the majorization func^c is equivalent to minimization of $g_0(\beta_0, \beta_1, \ldots \beta_p)$.

In IRWLS, we minimise $g_0(\beta_0, \beta_1, \ldots \beta_p)$.

LS - Regression

Here we need to minimize

$$\sum (y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - \cdots - \beta_p x_{pi})^2$$

The normal eq^{ns} are

$$\sum (y_i - \beta_0 - \cdots - \beta_p x_{pi}) = 0 \quad \Big| \quad \sum e_i = 0 \quad \Big| \quad \sum y_i = n\beta_0 + \beta_1 \sum x_{1i} + \cdots + \beta_p \sum x_{pi}$$

$$\sum (y_i - \beta_0 - \cdots - \beta_p x_{pi}) x_{1i} = 0 \quad \Big| \quad \sum e_i x_{1i} = 0 \quad \Big| \quad \sum x_{1i} y_i = \beta_0 \sum x_{1i} + \beta_1 \sum x_{1i}^2 + \cdots + \beta_p \sum x_{pi} x_{1i}$$

$$\sum (y_i - \beta_0 - \cdots - \beta_p x_{pi}) x_{2i} = 0 \quad \Big| \quad \sum e_i x_{2i} = 0 \quad \Big| \quad \sum x_{2i} y_i = \beta_0 \sum x_{2i} + \beta_1 \sum x_{1i} x_{2i} + \cdots + \beta_p \sum x_{pi} x_{2i}$$

$$\vdots$$

$$\sum (y_i - \beta_0 - \cdots - \beta_p x_{pi}) x_{pi} = 0 \quad \Big| \quad \sum e_i x_{pi} = 0 \quad \Big| \quad \sum x_{pi} y_i = \beta_0 \sum x_{pi} + \beta_1 \sum x_{1i} x_{pi} + \cdots + \beta_p \sum x_{pi}^2$$

Data set

$$\begin{array}{l} x_{01} = \\ x_{02} = \\ \vdots \\ x_{0n} = \end{array} \begin{pmatrix} 1 & x_{11} & x_{21} & \cdots & x_{p1} \\ 1 & x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{1n} & x_{2n} & \cdots & x_{pn} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = Y$$

$= X, y$ obs. Matrix

Data Matrix

$$X^T X = \begin{pmatrix} n & \sum x_{1i} & \cdots & \sum x_{pi} \\ \sum x_{1i} & \sum x_{1i}^2 & \cdots & \sum x_{pi} x_{1i} \\ \vdots & & & \\ \sum x_{pi} & \sum x_{1i} x_{pi} & \cdots & \sum x_{pi}^2 \end{pmatrix}$$

$$\begin{pmatrix} n & \sum x_{1i} & \sum x_{2i} & \cdots & \sum x_{pi} \\ \sum x_{1i} & \sum x_{1i}^2 & \sum x_{1i} x_{2i} & \cdots & \sum x_{1i} x_{pi} \\ \vdots & & & & \\ \sum x_{pi} & \sum x_{1i} x_{pi} & \sum x_{2i} x_{pi} & \cdots & \sum x_{pi}^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum y_i x_{1i} \\ \vdots \\ \sum y_i x_{pi} \end{pmatrix}$$

$$X^T X \beta = X^T Y$$

So in LS Regression we need to solve

$$\boxed{X^T X \beta = X^T Y}$$

$$\beta_{LS} = (X^T X)^{-1} X^T Y$$

$\text{rank}(X^T x) = p+1$ ← If rank is $p+1$, then it will be solvable.

Now to show $\text{rank}(X) = \text{rank}(X^T X)$

$$\dim(N(X)) = \#\text{ of col} - \dim[\text{Col}(X)]$$
$$= \#\text{ of col} - \ell(X)$$

Let $v \in N(X)$ $\quad \therefore Xv = 0 \Rightarrow (X^t X)v = 0$

$\therefore v \in N(X^t X) \Rightarrow N(X) \subseteq N(X^t X)$

and $v \in N(X^t X)$ $\quad X^t X v = 0 \Rightarrow v^T X^t X v = 0$

$\Rightarrow \|Xv\|^2 = 0$

$\Rightarrow Xv = 0$

$\therefore v \in N(X) \Rightarrow N(X^t X) \subseteq N(X)$

$\therefore \underline{\text{rank}(X)}_{p+1} = \text{rank}(X^T X)_{p+1}$

x_1 x_2 --- x_p	y	$\hat{y} = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi}$
	y_1	$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \dots + \beta_{1p} x$
	y_2	
	y_3	
	$\vdots$	
	y_n	$\hat{y}_n = \beta_0 + \beta_1 x_{1n} + \dots + \beta_p x_{pn}$

In LS Reg, we need to minimize,

$$\sum (y_i - \hat{y}_i)^2 = (y_1 - \hat{y}_1)^2 + \dots + (y_n - \hat{y}_n)^2$$
$$= \|y - \hat{y}\|^2 = (y - \hat{y})^T (y - \hat{y})$$

$$\hat{y} = \begin{pmatrix} X \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} = X\beta$$

In LS Regression, we minimize

$$\|y - X\beta\|^2 = (y - X\beta)^T (y - X\beta)$$

$$g(\beta) = (y - X\beta)^T (y - X\beta)$$
$$= y^T y - \beta^T X^T y - y^T X \beta + \beta^T X^T X \beta$$

$$\nabla g(\beta) = -X^T y - X^T y + 2X^T X \beta$$

$$\nabla g(\beta) = 0 \Rightarrow X^T X \beta = X^T Y \Rightarrow \boxed{\beta = (X^T X)^{-1} X^T Y}$$

(i) $Var(y) = Var(\hat{y}) + Var(e)$

(ii) $Cov(\hat{y}, e) = 0$ (iii) $Cov(\hat{y}, y) = Var(\hat{y})$

$$\Sigma \hat{y} e_i = \Sigma(\hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \dots + \hat{\beta}_p x_{pi}) e_i$$
$$\hat{\beta}_0 \Sigma e_i + \hat{\beta}_1 \Sigma x_{1i} e_i + \dots + \hat{\beta}_p \Sigma x_{pi} e_i = 0.$$

$\Sigma e_i = 0$.

$$Cov(\hat{y}, e) = \frac{1}{n} \underbrace{\Sigma \hat{y}_i e_i}_{0} - \bar{\hat{y}} \underbrace{\bar{e}}_{=0} = 0 \text{ --- (ii)}$$

$$y = \hat{y} + e \Rightarrow Var(y) = Var(\hat{y}) + Var(e) + \cancel{Cov(\hat{y}, e)}^{0}$$

$$\therefore Var(y) = Var(\hat{y}) + Var(e) \text{ --- (i)}$$

$$Cov(\hat{y}, y) = Cov(\hat{y}, \hat{y} + e) = Var(\hat{y}) + \cancel{Cov(\hat{y}, e)}^{0}$$
$$= Var(\hat{y}) \text{ --- (iii)}.$$

Multiple $R^2 = \frac{Var(\hat{y})}{Var(y)} = Corr^2(\hat{y}, y)$

Proof :- $Corr(y, \hat{y}) = \frac{Cov(y, \hat{y})}{\sqrt{Var(y)}\sqrt{Var(\hat{y})}} = \sqrt{\frac{Var(\hat{y})}{Var(y)}}$

In the case of simple linear regression

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$Corr^2(y, \hat{y}) = Cor^2(y, \hat{\beta}_0 + \hat{\beta}_1 x) = Corr^2(y, \hat{\beta}_1 x)$$
$$= Corr^2(y, x).$$

$Corr(y, a_1 x_1 + \dots + a_p x_p)$

$$= Corr[y - \bar{y}, a_1(x_1 - \bar{x}_1) + \dots + a_p(x_p - \bar{x}_p)]$$
$$y^*, \quad x_1^* + \dots + x_p^*$$

Mean $y^* = 0$, Mean $x_1^* = 0$ $\therefore$ Regression line passes through mean which is origin $(0, 0, \dots, 0)$

$\therefore \beta_0 = 0$.

If we regress y^* on $x_1^*, \dots x_p^*$ we get

$$\begin{pmatrix}\beta_1\\ \beta_2\\ \vdots\\ \beta_p\end{pmatrix} = (X^TX)^{-1}X^TY$$

where

$$X = \begin{pmatrix} x_{11}-\bar{x}_1 & x_{21}-\bar{x}_2 & & x_{p1}-\bar{x}_p \\ x_{12}-\bar{x}_1 & x_{21}-\bar{x}_2 & \cdots & x_{p2}-\bar{x}_p \\ \vdots & \vdots & & \vdots \\ x_{1n}-\bar{x}_1 & x_{2n}-\bar{x}_2 & & x_{pn}-\bar{x}_p \end{pmatrix}$$

$$X^TX = n\begin{pmatrix} Var(X_1) & Cov(X_1,X_2) & \cdots & Cov(X_1,X_p) \\ Cov(X_2,X_1) & Var(X_2) & - & Cov(X_2,X_p) \\ \vdots & \vdots & & \vdots \\ Cov(X_p,X_1) & Cov(X_p,X_2) & - & Var(X_p) \end{pmatrix}$$

$$= \underline{n S_{XX}}$$

$$X^TY = n\begin{pmatrix} Cov(X_1, Y) \\ Cov(X_2, Y) \\ \vdots \\ Cov(X_p, Y)\end{pmatrix} = \underline{n S_{XY}}$$

$$y = \begin{pmatrix} y_1 - \bar{y} \\ \vdots \\ y_n - \bar{y}\end{pmatrix}$$

$$\beta = S_{XX}^{-1} S_{XY} \quad [\because \beta = \underline{(X^TX)^{-1}}\, \underline{X^TY}]$$

$$corr(y-\bar{y},\ a_1(x_1-\bar{x}) + \dots + a_p(x_p - \bar{x}_p))$$

$$= Corr(y,\ a_1x_1 + \dots + a_px_p)$$ [∵ origin change doesn't affect correlation]

$$= \frac{Cov(y,\ a_1x_1 + \dots + a_px_p)}{\sqrt{Var(y)\, \underbrace{Var(a_1x_1 + \dots + a_px_p)}_{a^T S_{xx} a}}} = \frac{a^T S_{XY}}{\sqrt{(a^T S_{xx} a)\, S_y^2}}$$

Let us look at the square of the quantity.

$$\frac{(a^T S_{XY})^2}{S_y^2\,(a^T S_{XX} a)}$$

Eigen value decomp

$$= \frac{(a^T S_{xx} a)(a^T S_{xx} a)}{a^T S_{xx} a}$$

$$(a^T S_{XY})^2 = (\underbrace{a^T S_{XX}^{1/2}}_{u'}\ \underbrace{S_{XX}^{-1/2} S_{XY}}_{v})^2$$

$$(u'v) \le u'u \quad v'v \qquad (\Sigma u_i v_i)^2 \le (\Sigma u_i^2)(\Sigma v_i^2)$$

$$\frac{(a^T S_{XY})^2}{S_y^2\,(a^T S_{xx} a)} \le \frac{(a^T S_{XX}^{1/2} S_{XX}^{1/2} a)(S_{XY}^T S_{XX}^{-1/2} S_{XX}^{-1/2} S_{XY})}{S_y^2\,(a^T S_{xx} a)}$$

$\therefore$ Max^m value is

$$= \frac{\cancel{a^T S_{xx} a} \; S_{xy}^T S_{xx}^{-1} S_{xy}}{S_y^2 \; \cancel{a^T S_{xx} a}}$$

$$= \frac{1}{n} \cdot \frac{y^T X (X^T X)^{-1} X^T y}{S_Y^2}$$

$n S_{xx} = X^T X$

$X'y = n S_{xy}$

$y'X = n S_{xy}^T$

$\therefore$ Max^m value of correlation is

$$\frac{\sqrt{(y^T X (X^T X)^{-1} X^T Y)/n}}{S_Y}$$

[$\because$ we have squared]

Here the "$=$" holds (i.e. the correlation reaches the upper bound) when

$u \propto v$

$S_{xx}^{1/2} a \propto S_{xx}^{-1/2} S_{xy}$

$\Rightarrow a \propto S_{xx}^{-1} S_{xy}$

$= (X^T X)^{-1} (X'Y) = \beta$

$\therefore a \propto \beta \leftarrow$ max^m when $\underline{a \propto \beta}$

Now we need to show $\sqrt{((y^T X)(X^T X)^{-1} X^T Y)/n}$ is $\sqrt{Var(\hat{y})}$

$\hat{Y} = X\hat{\beta} = \underbrace{X(X^T X)^{-1} X^T}_{P_x} Y = P_x Y$

$\downarrow$

Projection matrix

$\hat{Y}^T \hat{Y} = Y^T P_x P_x Y$ [$\because P_x \to$ symmetric]

$= Y^T X (X^T X)^{-1} X^T X (X^T X)^{-1} X^T Y$

$\boxed{= Y^T X (X^T X)^{-1} X^T Y}$

$$Var(\hat{Y}) = \frac{\hat{Y}^T \hat{Y}}{n} \quad [\because \bar{\hat{Y}} = 0] \text{ where of our data set.}$$

$$\Rightarrow Var(\hat{Y}) = \frac{Y^T X (X^T X)^{-1} X^T Y}{n}$$

$\therefore$ our quantitity i.e.

$$Corr(y, a_1 x_1 + a_2 x_2 + \dots + a_p x_p) \text{ and it is max}^m \text{ where } a_i \propto \beta_i$$

$$= \sqrt{\frac{Var(\hat{Y})}{Var(y)}} = R$$

So the maximization correlation turns out to be R, multiple correlation coefficient $= r_{y \cdot 123 \dots p}$

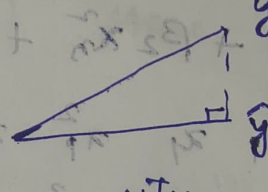
$R \geq Corr(Y, x_i)$ for any i $\rightarrow$ Here we can say by taking and $a_i = 1$ and all other $a_i = 0$ then we will get $Corr(y, x_i)$

max when all $a_i \propto \beta_i$

Check that — (i) $P_x^2 = P_x$, (ii) $(I - P_x)^2 = I - P_x$

(iii) $(I - P_x) P_x = 0$.

X_1	X_2	--	X_p	Y
				y_1
				⋮
				y_n
				y



$e = y - \hat{y}$

$$y^T y = \hat{y}^T \hat{y} + e^T e \quad [\because \text{ we know } \sum \hat{y}_i e_i = 0]$$

$$\Rightarrow \sum y_i^2 = \sum \hat{y}_i^2 + \sum e_i^2$$

$$\Rightarrow Var(y) = Var(\hat{y}) + Var(e)$$

(Proved)

Y, x_1, x_2, $\hat{Y} = \beta_1 x_1 + \beta_2 x_2$

Y on x_1

$\hat{Y} = \beta_1^* x_1$

$Y^* = Y - \beta_1^* x_1$

Y^* on x_2 $\rightarrow$ will not be the same as β_2

Reg Y on x_1

$\hat{Y} = \beta_1^* x_1$

$\Rightarrow Y^* = Y - \beta_1^* x_1$

Reg x_2 on x_1

$\hat{x}_2 = \alpha^* x_1$

$x_2^* = x_2 - \alpha^* x_1$

what if we reg. Y^* on x_2^*

It will be same as β_2

When we do multiple linear regression and fit a model

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

these β coefficients are called partial regression coefficients.

$$\beta_1 = \beta_{Y1.23\dots p}, \quad \beta_2 = \beta_{Y2.134\dots p}$$

$$\vdots$$

$$\beta_p = \beta_{Yp.123\dots p-1}$$

We will regress Y on $X_1, X_2, \dots X_{p-1}$ — (i)

and then we will regress X_p on $X_1, X_2, \dots X_{p-1}$ — (ii)

and then regress (i) & (ii) we will get β_p.

---

① $\hat{Y} = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3$

Data set

X	Y
x_1	y_1
$\vdots$	
x_n	y_n

$$\hat{y}_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_1^3$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_2 + \beta_2 x_2^2 + \beta_3 x_2^3$$

$$\vdots$$

$$\hat{y}_n = \beta_0 + \beta_1 x_n + \beta_2 x_n^2 + \beta_n x_n^3$$

$$\hat{Y} = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 \end{pmatrix}}_{X \text{ matrix}} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}}_{\beta}$$

② $Y = \beta_0 X^{\beta_1}$

$$\log Y = \log \beta_0 + \beta_1 \log X$$

$$Y^* = \beta_0^* + \beta_1^* X^*$$

X	Y	$X^* = \log X$	$Y^* = \log Y$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

2nd method

$\hat{Y} = \beta_0 x^{\beta_1}$

$y = \underbrace{\beta_0 x^{\beta_1}}_{f(x)} + \varepsilon$

$\sum [y_i - f(x_i)]^2$

$\sum (y_i - \beta_0 x_i^{\beta_1})^2$

We can start with $\beta_0^{(0)}$ and $\beta_1^{(0)}$ and update the parameters

1st Taylors Expansion

$$f(x,y) = f(x_0, y_0) + \frac{1}{\lfloor 1} \left[(x-x_0) \frac{\partial f}{\partial x}\Big|_{x=x_0} + (y-y_0) \frac{\partial f}{\partial y}\Big|_{y=y_0} \right] + \frac{1}{\lfloor 2} \left[(x-x_0)^2 \frac{\partial^2 f}{\partial x^2}\Big|_{x=x_0} + (y-y_0)^2 \frac{\partial^2 f}{\partial y^2}\Big|_{y=y_0} + 2(x-x_0)(y-y_0) \frac{\partial^2 f}{\partial x \partial y}\Big|_{\substack{x=x_0 \\ y=y_0}} \right] + \ldots$$

$$f(x,y) \simeq f(x_0, y_0) + (x-x_0) \frac{\partial f}{\partial x}\Big|_{x=x_0} + (y-y_0) \frac{\partial f}{\partial y}\Big|_{y=y_0}$$

$\sum (y_i - \beta_0 x_i^{\beta_1})^2 \rightarrow f(\beta_0, \beta_1) = \beta_0 x^{\beta_1}$

$$\simeq \sum \left[\underbrace{y_i - \{\beta_0^{(0)} x_i^{\beta_1^{(0)}}}_{z_i} + \underbrace{(\beta_0 - \beta_0^{(0)})}_{\alpha_0} x_i^{\beta_1^{(0)}} + \underbrace{(\beta_1 - \beta_1^{(0)})}_{\alpha_1} \beta_0^{(0)} x_i^{\beta_1^{(0)}} \ln x_i \} \right]^2$$

$= \sum (z_i - \alpha_0 w_i - \alpha_1 u_i)^2$ Here α_0, α_1 can be found out using L-S Regression of multi-variable.

where we get $\hat{\alpha}_0 = \hat{\beta}_0 - \beta_0^{(0)}$

and $\hat{\alpha}_1 = \hat{\beta}_1 - \beta_1^{(0)}$

From there we compute $\beta_0^{(1)} = \hat{\beta}_0 = \hat{\alpha}_0 + \beta^{(0)}$

and $\beta_1^{(1)} = \hat{\beta}_1 = \hat{\alpha}_1 + \beta_1^{(0)}$

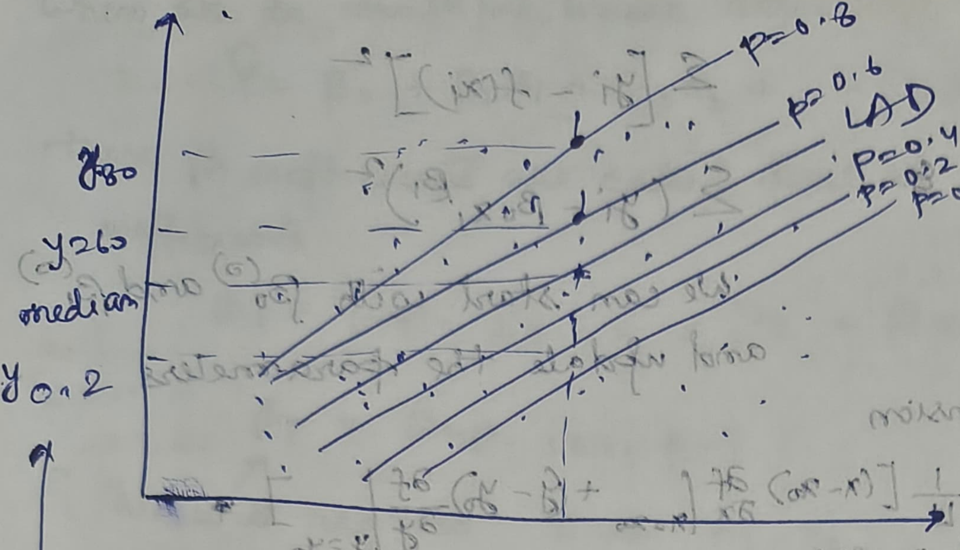
Quantile Regression.

$\sum (x_i - \theta)^2$, $\sum |x_i - \theta|$, $\sum |x_i - \theta| + (2p-1) \sum (x_i - \theta)$

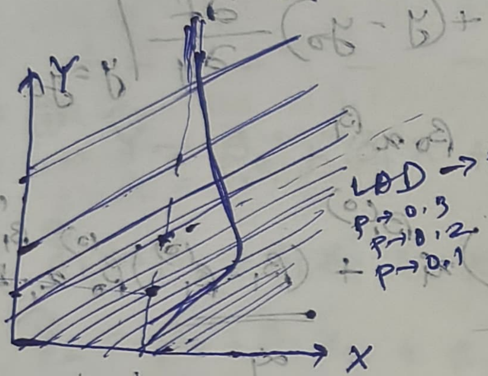
p^{th} quantile

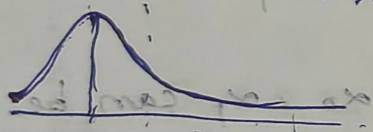
$\sum [y_i - \beta_0 - \beta_1 x_i]^2$ $\quad \sum |y_i - \beta_0 - \beta_1 x_i|$

IRWLS algo is used for quantile regression $\leftarrow$ $\sum |y_i - \beta_0 - \beta_1 x_i| + (2p-1) \sum (y_i - \beta_0 - \beta_1 x_i)$



Here we will get median, quantiles for given value of x



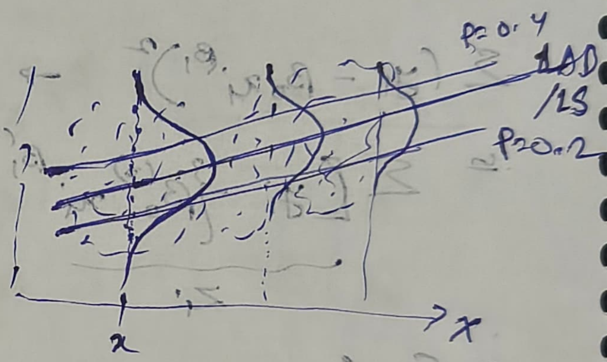


Right skewed.

e ~ distribution - skewed with const σ

lines will $||^l$ and gaps are unequal

Homo skedastic



Reg. Line: $E(Y \mid X = x)$

error ~ $N(0, \sigma)$

lines are $||^l$ and gaps are equal

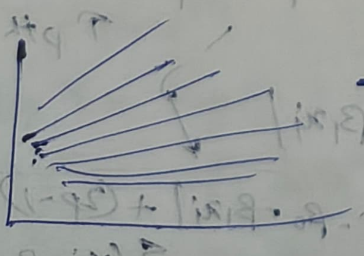
e variance = const

$y|x=x$ " = const

Line passing through mean/median of Normal dist will be LAD

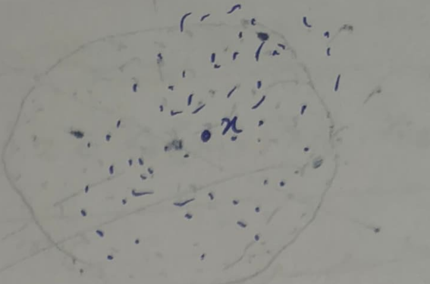
Non-Homoskedastic → Variance is variable

→ Non $||^l$ lines.



28/09/2025.

Mahalanobis Depth



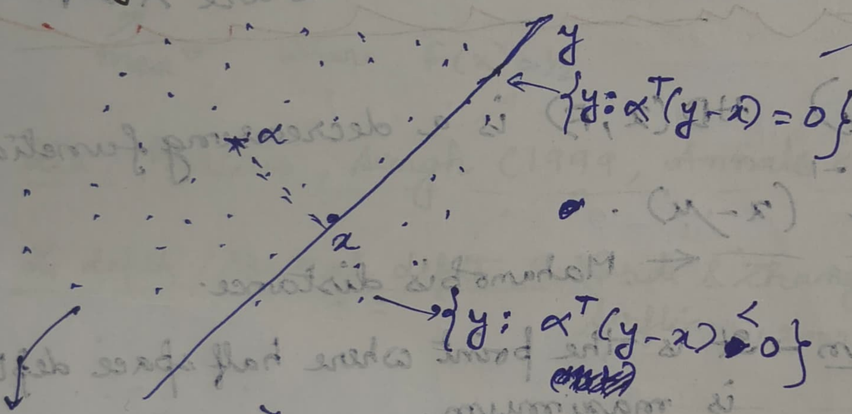
$$MD(x, \Omega_n) = \frac{1}{1 + (x - \bar{x})^T S^{-1} (x - \bar{x})}$$

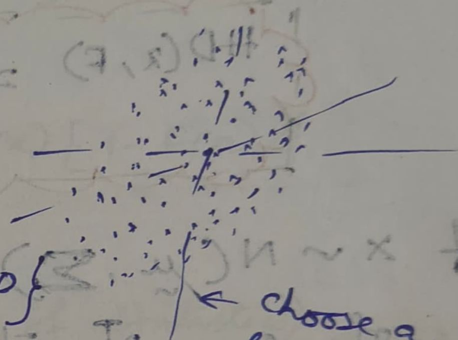
where $\Omega_n = \{x_1, x_2, \dots x_n\}$

where S is the sample dispersion (var-cov) matrix.

But this is not robust because for outliers both S and $\bar{x}$ gets affected.

Tukey Depth / Half space depth (1975)





choose a line for which one half space has min^m no. of obs. ~~and~~ at max^m.

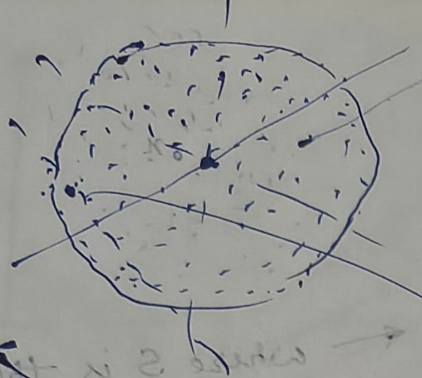
$$H^+ = \{y : \alpha^T(y - x) \geq 0\}$$

$$H^- = \{y : \alpha^T(y - x) \leq 0\}$$

$$\inf_{\alpha : \|\alpha\| = 1} \min \left\{ \frac{\sum_{i=1}^{n} \mathbb{1}\{\alpha^T(x_i - x) \geq 0\}, \sum_{i=1}^{n} \mathbb{1}\{\alpha^T(x_i - x) < 0\}}{n} \right\}$$

$HD(x, \Omega_n)$.

$$\Rightarrow HD(x, \Omega_n) = \inf_{\alpha : \|\alpha\| = 1} \frac{\sum_{i=1}^{n} \mathbb{1}\{\alpha^T(x_i - x) \geq 0\}}{n}$$



← For observation nearing centre

$HD = \frac{1}{2}$

For observation towards outside

$HD << \frac{1}{2}$

For outliers $HD = 0$.

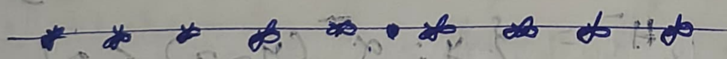
Now using probability distribution,

$$\boxed{HD(x, F) = \inf_{\alpha : \|\alpha\| = 1} P(\alpha'(X - x) \geq 0) \quad \text{where } X \sim F}$$

If $X \sim N(\mu, \Sigma)$, $HD(x; F)$ is a decreasing function of $\underbrace{(x-\mu)^T \Sigma^{-1} (x-\mu)}$. ← Mahanobis distance.

Half space Median → It is the point where half space depth is maximum.

For single variate data cloud



← $\alpha = \pm 1$

$$\cancel{HD(x, \Omega_n) = \inf_{\alpha} \left[\sum \mathbb{1}\{x_i \geq x\} \right]}$$

$\alpha'(x_i - x) \geq 0$

$\Rightarrow x_i \geq x$

$$HD(x, \Omega_n) = \inf_{\alpha} \left[\frac{\sum \mathbb{1}\{\alpha^T (x_i - x) \geq 0\}}{n} \right]$$

$$= \min \left[\frac{\sum \mathbb{1}\, x_i \geq x}{n}, \frac{\sum \mathbb{1}\, x_i \leq x}{n} \right]$$

This is max^m when x is median of x_i's.

Liu (1990): Simplicial depth:

$$\boxed{SD(x, \Omega_n) = \frac{\sum_{1 \le i_1 < i_2 < i_3 \le n} \mathbb{1}\{x \in \Delta(X_{i_1}, X_{i_2}, X_{i_3})\}}{{}^nC_3}}$$

$$SD(x, F) = P[x \in \Delta(X_1, X_2, X_3)]$$

where $X_1, X_2, X_3 \overset{iid}{\sim} F$

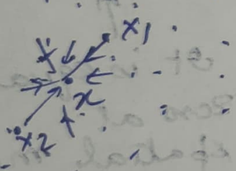
For single variate dist.

$P[x \in \text{Interval}(X_1, X_2)] = 2F(x)[1 - F(x)]$

max^m where $F(x) = 1/2$

Liu, Parelius, Singh (1999, Annals of Statistics) ← Paper for studying depths.

L_1 depth / Spatial depth (Vardi & Zhang, 2000
Serfling, 2002)



$$SPD(x, \Omega_n) = 1 - \left\| \frac{1}{n} \sum_{i=1}^{n} \frac{x - X_i}{\|x - X_i\|} \right\|$$

$$SPD(x, F) = 1 - \left\| E\left(\frac{x - X}{\|x - X\|} \right) \right\| \quad X \sim F$$

In 1 direction

$$\frac{x - X}{\|x - X\|} = 1 \text{ if } X < x$$
$$= -1 \text{ if } X > x$$

$$E\left(\frac{x - X}{\|x - X\|}\right) = 1 \cdot P(X < x) + (-1) P(X > x)$$
$$= F(x) - [1 - F(x)] = 2F(x) - 1$$

$SPD(x, F) = 1 - |2F(x) - 1|$ ← For ~~median~~ SPD max^m

$\Rightarrow F(x) = \frac{1}{2}$ ← median

For higher dimensions,

① $\sum \|\underset{\sim}{x}_i - \underset{\sim}{\theta}\|^2$ $\qquad x_1, x_2, \ldots x_n$

$= \sum (x_i^{(1)} - \theta^{(1)})^2$

$+ \sum (x_i^{(2)} - \theta^{(2)})^2 + \cdots + \sum (x_i^{(p)} - \theta^{(p)})^2$

gets minimized at co-ordinate wise ~~median~~ mean

② $\sum \|\underset{\sim}{x}_i - \underset{\sim}{\theta}\|_1$ $\qquad a = (a_1, a_2, \ldots a_p)$

gets minimized at co-ordinate wise median $\qquad \|\underset{\sim}{a}\|_1 = |a_1| + |a_2| + \cdots + |a_p|$

③ $\psi(\theta) = \sum \|x_i - \theta\|_2$ $\qquad \|a\|_2 = \sqrt{a_1^2 + a_2^2 + \cdots + a_p^2}$

$$\nabla \psi(\theta) = -\sum_{i=1}^{n} \begin{pmatrix} \frac{x_i^{(1)} - \theta^{(1)}}{\|x_i - \theta\|} \\ \frac{x_i^{(2)} - \theta^{(2)}}{\|x_i - \theta\|} \\ \vdots \\ \frac{x_i^{(p)} - \theta^{(p)}}{\|x_i - \theta\|} \end{pmatrix} = -\sum_{i=1}^{n} \frac{x_i - \theta}{\|x_i - \theta\|}$$

It is equal to zero at spatial median

$\therefore \psi(\theta)$ is getting minimized at spatial median.

④ $\psi(\theta) = \sum |x_i - \theta| + u \sum (x_i - \theta)$

$\xrightarrow{\min}$ u-th spatial quantile $\qquad u = 2p - 1$, $(-1, 1)$ | p^{th} quantile $(0,1)$

For higher dimension
u-th spatial quantile will minimize of

$$\sum \|x_i - \theta\|_2 + \sum_{i=1}^{n} u^T (x_i - \theta)$$

← useful for drawing q-q plot for higher dimension (Sinica, 1998) John Marden

26/09/2025.

Regression

$$Y = \underbrace{f(X_1, X_2 \ldots X_p)}_{\text{signal}} + \underbrace{\varepsilon}_{\text{noise}}$$

In the case of linear model, we have

$$f(X_1, X_2 - X_p) = \beta_0 + \beta_1 X + \beta_2 X + \cdots + \beta_p X_p$$

~~Here~~ we ~~conclude~~ consider the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_p X_{pi} + \varepsilon_i$$

$$i = 1, 2, \ldots n$$

Let us assume that the X_i's are non-stochastic and ε_i's are independent with $E(\varepsilon_i) = 0$ and $Var(\varepsilon_i) = \sigma^2$ for all i

$$E(Y_i) = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_p X_{2i}$$

$$Var(Y_i) = \sigma^2 \quad \forall\ i = 1, 2, \ldots n$$

If we consider X_i's to be stochastic, then we have

$$E(Y_i \mid X_{1i} \ldots X_{pi}) = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_p X_{pi}$$

$$Var(Y_i \mid X_i \ldots X_{pi}) = \sigma^2$$ under the assumption that ε_i's and X_i's are uncorrelated

This model can also be expressed as

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$

$$E(\underset{\sim}{Y}) = X\underset{\sim}{\beta}$$

$$Disp(\underset{\sim}{Y}) = \sigma^2 \mathbb{I}$$

$$Y_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{21} + \cdots + \beta_p X_{p1} + \varepsilon_1$$

$$Y_2 = \beta_0 + \beta_1 X_{21} + \beta_2 X_{22} + \cdots + \beta_p X_{p2} + \varepsilon_2$$

$$\vdots$$

$$Y_n = \beta_0 + \beta_1 X_{n1} + \beta_2 X_{n2} + \cdots + \beta_p X_{pn} + \varepsilon_n$$

$$E(\underset{\sim}{\varepsilon}) = \begin{bmatrix} E(\varepsilon_1) \\ E(\varepsilon_2) \\ \vdots \\ E(\varepsilon_n) \end{bmatrix} = \underset{\sim}{0}$$

$$Var(\underset{\sim}{\varepsilon}) = Disp(\underset{\sim}{\varepsilon})$$

$$= \begin{pmatrix} Var(\varepsilon_1) & Cov(\varepsilon_1, \varepsilon_2) & \cdots & Cov(\varepsilon_1, \varepsilon_n) \\ Cov(\varepsilon_1, \varepsilon_2) & Var(\varepsilon_2) & \cdots & Cov(\varepsilon_2, \varepsilon_n) \\ \vdots & \vdots & & \\ Cov(\varepsilon_n, \varepsilon_1) & Cov(\varepsilon_n, \varepsilon_2) & \cdots & Var(\varepsilon_n) \end{pmatrix} = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix}$$

[$\because \varepsilon_1, \varepsilon_2$ are independent $\therefore Cov = 0$]

Now we know,

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$E(\hat{\beta}) = (X'X)^{-1} X' \underline{E(Y)}$$

$$\Rightarrow E(\hat{\beta}) = (X'X)^{-1} X' X \beta = \beta$$

$\therefore E(\hat{\beta}) = \beta$ and $\hat{\beta}$ is an unbiased estimastor of B

$$Var(\hat{\beta}) = Disp[\underbrace{(X'X)^{-1} X}_{A} Y]$$

$$= A \cdot Disp(Y) A'$$

$Disp(X) = \Sigma$

$Var(a'X) = a' \Sigma a$

$\therefore$ $Disp(\hat{\beta}) = (X'X)^{-1}X' \sigma^2 \mathbb{I} \, X(X'X)^{-1}$

$= \sigma^2 (X'X)^{-1} \underline{X'X} (\underline{X'X})^{-1} = \sigma^2 (X'X)^{-1}$

$\therefore$ We have seen, $\boxed{E(\hat{\beta}) = \beta \text{ and } Disp(\hat{\beta}) = \sigma^2 (X'X)^{-1}}$

For Simple Linear Regression, $Y = \beta_0 + \beta_1 X + \varepsilon$, ε_i's are independent

$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ $\qquad \left.\begin{matrix} E(\varepsilon_i) = 0 \\ Var(\varepsilon_i) = \sigma^2 \end{matrix}\right\} \forall i$

$\hat{\beta}_1 = \dfrac{\Sigma (x_i - \bar{x})(Y_i - \bar{Y})}{\Sigma (x_i - \bar{x})^2}$

$E(\hat{\beta}_0), E(\hat{\beta}_1), Var(\hat{\beta}_0), Var(\hat{\beta}_1), Cov(\hat{\beta}_0, \hat{\beta}_1)$

$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}$ $\quad X'X = \begin{pmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{pmatrix}$ $\quad (X'X)^{-1} = \dfrac{1}{n\Sigma x_i^2 - (\Sigma x_i)^2} \begin{pmatrix} \Sigma x_i^2 & -\Sigma x_i \\ -\Sigma x_i & n \end{pmatrix}$

$E(\hat{\beta}) = \beta \Rightarrow E\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$

$Disp(\hat{\beta}) = \sigma^2 (X'X)^{-1} = \dfrac{\sigma^2 \cdot n}{n\Sigma x_i^2 - (\Sigma x_i)^2} \begin{pmatrix} \frac{\Sigma x_i^2}{n} & -\frac{\Sigma x_i}{n} \\ -\frac{\Sigma x_i}{n} & \frac{n}{n} \end{pmatrix}$

$= \dfrac{\sigma^2}{n\left[\frac{1}{n}\Sigma x_i^2 - \bar{x}^2\right]} \begin{pmatrix} \frac{1}{n}\Sigma x_i^2 & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix} = \dfrac{\sigma^2}{n \cdot \frac{1}{n} \Sigma (x_i - \bar{x})^2} \begin{pmatrix} \frac{1}{n}\Sigma x_i^2 & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix}$

$= \dfrac{\sigma^2}{\Sigma (x_i - \bar{x})^2} \begin{pmatrix} \frac{1}{n}\Sigma x_i^2 & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix}$

$\therefore$ $Var(\hat{\beta}_0) = \dfrac{\sigma^2}{\Sigma (x_i - \bar{x})^2} \cdot \dfrac{1}{n}\Sigma x_i^2$, $\quad Var(\hat{\beta}_1) = \dfrac{\sigma^2}{\Sigma (x_i - \bar{x})^2}$

$Cov(\hat{\beta}_0, \hat{\beta}_1) = \dfrac{\sigma^2}{\Sigma (x_i - \bar{x})^2} (-\bar{x})$.

Regression Problem $\rightarrow$

Y	X_1	X_2
|	|	|

Assuming
$\bar{Y} = \bar{X}_1 = \bar{X}_2 = 0$ (centred)
$S_1^2 = S_y^2 = S_{x_2}^2 = 1$ (standardised)

Consider a model

$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$

where ε_i's are independent $\left.\begin{matrix} E(\varepsilon_i) = 0 \\ Var(\varepsilon_i) = \sigma^2 \end{matrix}\right\} \forall i$ $\quad \therefore X = \begin{pmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \\ \vdots & \vdots \\ x_{1n} & x_{2n} \end{pmatrix}$

$X'X = \begin{pmatrix} \Sigma x_{1i}^2 & \Sigma x_{1i} x_{2i} \\ \Sigma x_{1i} x_{2i} & \Sigma x_{2i}^2 \end{pmatrix}$

We can write $\sum x_{1i}^2 = n \cdot \frac{1}{n} \sum x_{1i}^2$ $[\because \bar{x}^2 = 0]$

$= n\left(\frac{1}{n} \sum x_{1i}^2 - \bar{x}^2\right)$

$\therefore X'X = \begin{pmatrix} n & S_{x_1 x_2} \\ S_{x_1 x_2} & n \end{pmatrix}$ $= n \cdot \underline{\underline{S_{x_1}^2}}_{1} = n$

$X'X = n \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ where $r = Corr(X_1, X_2)$ ~~$(X'X)^{-1} = \frac{1}{1-r^2}$~~

$(X'X)^{-1} = \frac{1}{n(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}$ $X'Y = \begin{pmatrix} \sum X_{1i} Y_i \\ \sum X_{2i} Y_i \end{pmatrix} = n \begin{pmatrix} r_{YX_1} \\ r_{YX_2} \end{pmatrix}$

$\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = (X'X)^{-1} X'Y = \frac{1}{\cancel{n}(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix} \cdot \cancel{n} \begin{pmatrix} r_{YX_1} \\ r_{YX_2} \end{pmatrix}$

$= \frac{1}{1-r^2} \begin{pmatrix} r_{YX_1} - r \cdot r_{YX_2} \\ r_{YX_2} - r \cdot r_{YX_1} \end{pmatrix}$

~~Disp($\hat{\beta}$)~~ $Disp(\hat{\beta}) = \sigma^2 (X'X)^{-1} = \frac{\sigma^2}{n(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}$

- If X_1 and X_2 are highly correlated then

i) $Disp(\hat{\beta}) \to \infty$ and also ii) $\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} \to \infty$ $\because r^2 = 1$

iii) Sign of regression coefficient would be misleading $\to$ $r_{YX_1} = 0.8$, $r_{YX_2} = 0.3$, $r = 0.5$ $\beta_1 \to +ve$, $\beta_2 \to -ve$

Variance of $\hat{\beta}_1$ and $\hat{\beta}_2$ increases

Magnitude of $\hat{\beta}_1$ and $\hat{\beta}_2$ increases

It will say that Y and X_2 are -vely correlated But actually $r_{YX_2} = 0.3$ +vely correlated

Problem of Multi Collinearity

near singularity of $X'X$ matrix $\to$ sum of ~~Some~~ λ close to 0

Let $\lambda_1 \geq \lambda_2 \geq \cdots \lambda_p \geq 0$ be the eigen values of $X'X$

then $(X'X)^{-1}$ $\underbrace{\overbrace{\frac{1}{\lambda_p} \geq \frac{1}{\lambda_{p-1}}}^{\text{close to zero}}}_{\text{large}} \cdots \geq \frac{1}{\lambda_2} \geq \frac{1}{\lambda_1} \geq 0$

$Var(\hat{\beta}_1) + Var(\hat{\beta}_2) + \cdots + Var(\hat{\beta}_p)$ We know $Disp(\hat{\beta}) = \sigma^2 (X'X)^{-1}$

$= \sigma^2 \underbrace{trace\left[(X'X)^{-1}\right]}_{\text{sum of eigen values}} = \sigma^2 \sum_{i=1}^{p} \frac{1}{\lambda_i}$ $\to$ this will be large in case of multi collinearity.

$$E[\|\hat\beta-\beta\|^2] = E[(\hat\beta_1-\beta_1)^2+(\hat\beta_2-\beta_2)^2+\cdots+(\hat\beta_p-\beta_p)^2]$$

$$= E[\hat\beta_1-\beta_1]^2 + E[\hat\beta_2-\beta_2]^2+\cdots+E[\hat\beta_p-\beta_p]^2$$

$$= Var(\hat\beta_1)+Var(\hat\beta_2)+\cdots+Var(\hat\beta_p)$$

$E[\hat\beta_i - E(\hat\beta_i)]^2 = Var(\hat\beta_i)$

$$= \sigma^2\left(\frac{1}{\lambda_1}+\cdots+\frac{1}{\lambda_p}\right)$$

To deal with this there are 2 ways → (i) variable deletion
(ii) adding λI to XX' so that sum of eigen values are not close to zero.

We are taking 2nd method for now.

$X'X$ has eigen values $\lambda_1 \geq \lambda_2 \cdots \geq \lambda_p$

$X'X+\lambda I$ " " " $\lambda_1+\lambda \geq \lambda_2+\lambda \cdots \geq \lambda_p+\lambda$ (where $\lambda \geq 0$)

$(X'X+\lambda I)^{-1}$ " " $\frac{1}{\lambda+\lambda_p} \geq \frac{1}{\lambda+\lambda_{p-1}} \geq \cdots \geq \frac{1}{\lambda_2+\lambda} \geq \frac{1}{\lambda_1+\lambda}$

So instead of taking $\hat\beta = (X'X)^{-1}X'Y$

one can use $\boxed{\hat\beta_R = (X'X+\lambda I)^{-1}X'Y}$ as a regularized estimate

Note that this regularization is done for reducing the variance of this estimate of the regression coefficient but

$$E(\hat\beta_R) = E[(X'X+\lambda I)^{-1}X'Y] = (X'X+\lambda I)^{-1}X'E(Y)$$
$$= (X'X+\lambda I)^{-1}X'X\beta \neq \beta$$

So it becomes a bias estimate.

The regularization parameter λ is often chosen using the cross-validation method.

Now in LS Regression we minimize $\|Y-X\beta\|^2$ to get $\hat\beta = (X'X)^{-1}X'Y$. ~~But we will now~~ ~~minimize~~ But in multi collinearity $\|\hat\beta\|^2$ blows up.

So in some cases, one can think of minimizing $\|Y-X\beta\|$ s.t. $\|\beta\|^2 \leq c$ for some $c>0$

This is equivalent to minimize $\|Y-X\beta\|^2+\lambda(\|\beta\|^2-c)$

where $\lambda > 0$ ← Lagrange's multiplier

For minimizing w.r.t β we get,

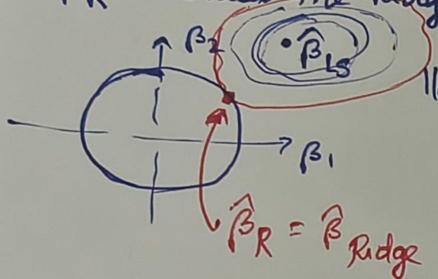
$$(Y-X\beta)^T(Y-X\beta)+\lambda(\beta^T\beta-c) = Y'Y-\beta^TX^TY-Y'X\beta+\beta'X'X\beta + \lambda\beta'\beta-\lambda c$$

Here the derivative will be

$$-2X'Y+2X'X\beta+2\lambda I\beta = 0$$
$$\Rightarrow \hat\beta_R = (X'X+\lambda I)^{-1}X'Y$$

$$\hat{\beta}_R = (X'X + \lambda I)^{-1} X'Y$$

This regularized regression method is known as <u>Ridge regression</u>. and $\hat{\beta}_R$ is called the Ridge estimate of the reg. coefficient.



$$\hat{\beta}_{LS} = (X'X)^{-1} X'Y$$

$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}_{LS}\|^2 + \underbrace{(\beta - \hat{\beta}_{LS})' X'X (\beta - \hat{\beta}_{LS})}_{\text{Eq}^n \text{ of ellipse}}$$

$$\|\beta\|^2 \le c \Rightarrow \beta_1^2 + \beta_2^2 + \cdots + \beta_p^2 \le c.$$